No. of Functional Features Included in the Solution

Date	25 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	5 Marks

Functional Features Overview:

The table lists every functional feature of the project with its implementation status and marks contribution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User registration with bcrypt hashing	Creates an account with validated name, email and password	authController / User model	Done	0.5
2	User login with JWT	Verifies credentials and returns a signed token	authController	Done	0.5
3	Protected profile endpoint	Returns the logged-in user's profile using the Bearer token	authMiddleware	Done	0.25
4	Request validation	Checks register, login, FAQ and AI payloads before processing	validationMiddleware	Done	0.25
5	Centralised error handling	Maps duplicate keys, validation and bad IDs to clean errors	errorMiddleware	Done	0.5
6	Create FAQ (protected)	Saves question, answer and category with the author reference	faqController / FAQ model	Done	0.5
7	View FAQs (public)	Lists all FAQs or fetches one by ID	faqController	Done	0.25
8	Update / delete own FAQ	Author-only modification of FAQ records	faqController	Done	0.5
9	Keyword FAQ search	Finds FAQs matching a search query	faqController	Done	0.5
10	Category classification	FAQ category limited to Technology, Education, Health, Banking, General	FAQ model	Done	0.25
11	AI answer (Gemini)	Returns an AI answer for a free-text question	aiController / geminiService	Done	0.5
12	AI FAQ generation (Gemini)	Generates a question, answer and category from a topic	aiController / geminiService	Done	0.5
13	Admin / editor role management	Admin panel and role-based permissions	-	Pending (planned)	0
14	Likes, answer feedback and tags	FAQ_LIKE, ANSWER_FEEDBACK and TAG entities from the ER design	-	Pending (planned)	0

Feature Summary:

Metric	Count / Value
Total Features Planned	14

Metric	Count / Value
Total Features Implemented	12
Core / Must-Have Features	10 (registration, login, validation, error handling, create/view/update-delete FAQ, search, AI answer, AI FAQ generation)
Additional / Nice-to-Have Features	2 implemented (profile endpoint, category classification); 2 planned
Features Tested & Verified	4 verified with screenshots in the Project Documentation (register, login, profile, AI answer); remaining endpoints are in the Postman collection

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	0	Headless API; Postman is the client
2	Backend / Logic	4	Validation, error handling, view FAQs, update/delete own FAQ
3	Database / Storage	2	User and FAQ Mongoose models, create FAQ
4	API / Integration	3	Keyword search, AI answer, AI FAQ generation
5	Security / Authentication	3	Registration with bcrypt, JWT login, protected profile